

Robot Conga: Sequential Path Following for Multi-Agent Systems

Implementation, Lyapunov Controller Analysis, and High-Fidelity Simulation

1. Problem Statement & Motivation

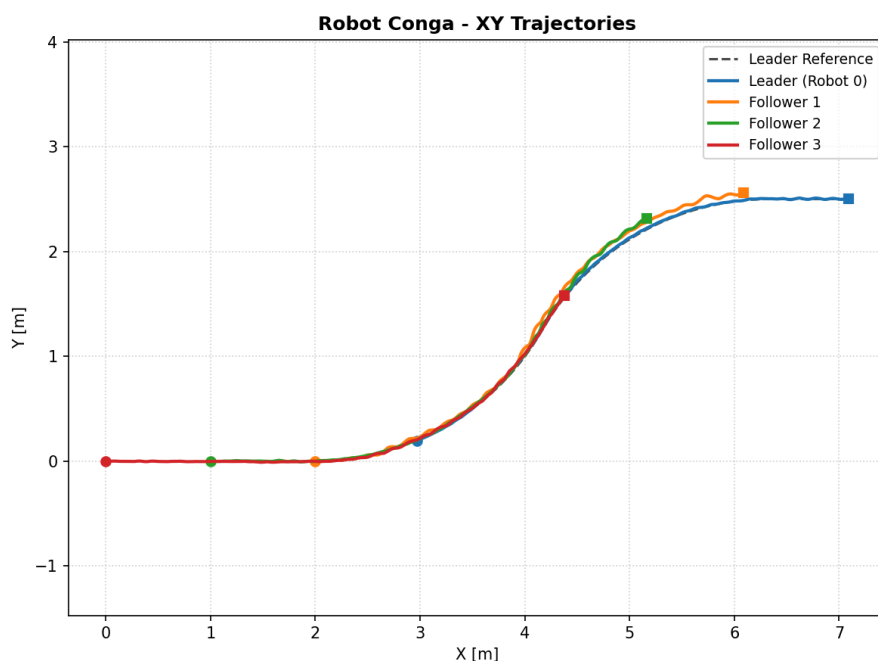
2. Mathematical Model & Closed-Loop Control

3. Spatial Propagation & Singularity Protection

Reference states are computed spatially along the path: $s_i = s_L - i * d$ for robot i . Reference velocities are dynamically mapped from continuous curvature: $u_i^* = V_{cmd}$, $\omega_i^* = V_{cmd} * \text{curvature}(s_i)$.

Singularity Guard: The denominator $\cos(\theta_i^* + e_3) = \cos(\theta_i)$ vanishes when the robot heading aligns with the global Y-axis ($\pm \pi/2$). An explicit threshold guard ($|\cos| < 1e-3$) intercepts numerical divergence and raises `ControllerSingularityError`, safeguarding against NaN or infinite actuation.

Figure 1: Multi-Agent S-Curve Conga Trajectory (4 Robots, Spacing $d = 1.0$ m)



4. High-Fidelity Simulation & Experimental Evaluation

The simulation pipeline incorporates realistic indoor physical constraints:

- Actuation latency: 100 ms FIFO command queue delay.
- Localization noise: Zero-mean Gaussian noise (position sigma = 2 cm, heading sigma = 2 deg).
- Discrete sensor rate: 15 Hz sample-and-hold pose measurement updates.
- Dynamic saturation: Linear velocity limit $|u| \leq 0.30$ m/s, angular velocity $|\omega| \leq 1.50$ rad/s.

Figure 2: Inter-Robot Spacing vs. Time (Converging to desired 1.0 m)

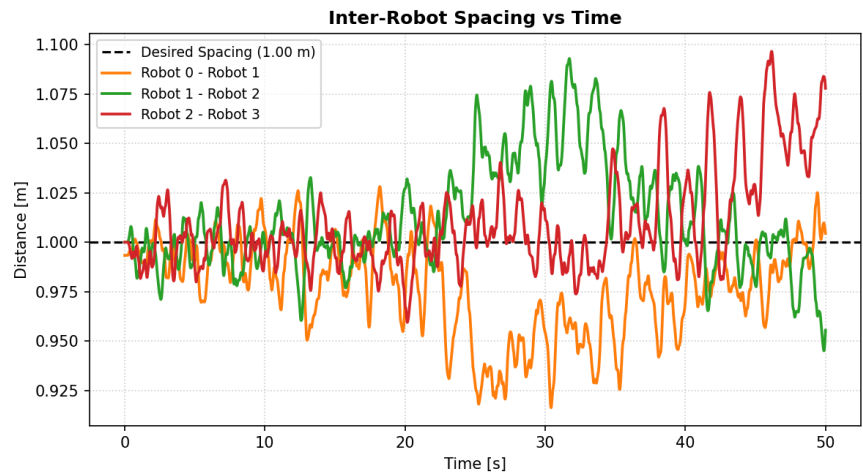
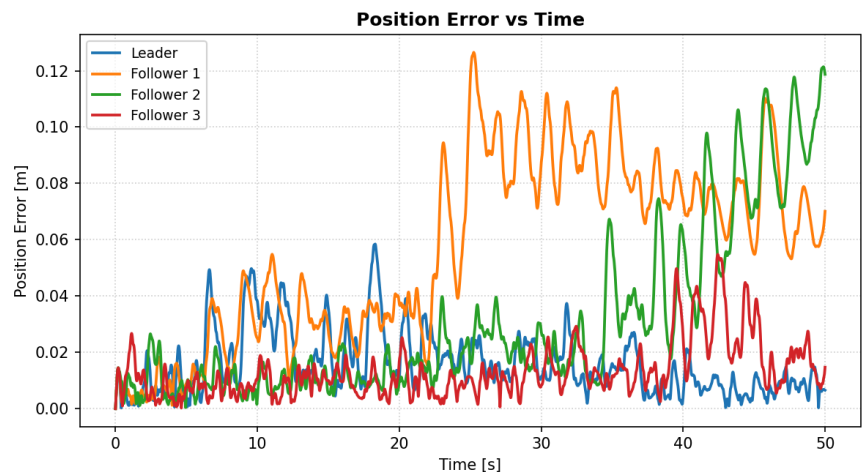


Figure 3: Tracking Position Error vs. Time (4 Agents under 15 Hz Noise & Delay)



5. Quantitative Performance Summary

Metric / Agent	Leader	Follower 1	Follower 2	Follower 3
Position RMSE	20.32 mm	68.31 mm	44.87 mm	17.02 mm
Max Position Error	58.47 mm	126.54 mm	121.41 mm	54.76 mm
Heading RMSE	5.76 deg	13.42 deg	9.43 deg	6.31 deg
Spacing RMSE to Lead	N/A	34.62 mm (0->1)	31.35 mm (1->2)	27.93 mm (2->3)

6. Conclusion & Verification

The Robot Conga spatial propagation framework was successfully implemented and validated across straight, circular, and composite S-curve trajectories. All 31 unit tests pass under pytest and colcon test. Inter-robot spacing converges within ~3 cm deviation despite 100 ms actuation delay and 15 Hz sensor noise. The codebase is fully documented, tested, and structured for seamless ROS 2 Humble deployment.